Aanya A. Gupta

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Education

Duke University Durham, NC
Electrical and Computer Engineering *Expected May 2029*
- **Relevant Coursework:** Data Structures/Computer Algorithms, Differential Equations, Chem/Biology, Engineering Design

Thomas Jefferson High School for Science and Technology (TJHSST) Alexandria, VA
Advanced Studies Diploma *June 2026*
- **Cumulative GPA:** 4.60 (W)
- **Relevant Coursework:** Statistics, Multivariable Calculus, Linear Algebra, AP Physics, Computer Vision, Mobile/Web Dev
- **Activities:** TJClimbing NGO (Founder), TJHSST TEKNOS Research Journal Club (Associate Editor-in-Chief), TJHSST Coding Lady Colonials Club (President), HackTJ Hackathon (President/Director)

Experience

Raytheon June 2026 – Aug. 2026
Software and Research Intern (Summer 2027 Return Offer Received) *Arlington, VA*
- Rebuilt a fragmented benchmarking pipeline into a modular, end-to-end evaluation framework in Python for a fraud-detection research program, reducing per-run setup from 4 hours to 10 minutes and making results reproducible across 5 data sources
- Benchmarked the robustness of existing anomaly-detection models against realistic synthetic event data, characterizing 3 failure modes where detectors missed suspicious activity; the test harness was adopted by a 35-person team and 3 customers

Agent-Based Modeling Research under Dr. Taylor Anderson May 2024 – Present
Research Intern *Fairfax, VA*
- Engineered an agent-based model of vaccine uptake across 5.5K synthetic agents linking social factors and health policy
- Pioneered a novel approach to small-area estimation using iterative proportional fitting, improving the accuracy of local health outcome predictions by 2% over established CDC benchmarks across 50+ counties
- Co-authored multiple publications and open-sourced model code and datasets for public health research and decision-making

Givology (Global Education 501(c)(3)) Sep. 2022 – Present
Head of Events and Special Projects *Fairfax, VA*
- Spearheaded 6 fundraising campaigns, raising over $18K to support education initiatives for children in developing countries
- Managed Facebook and LinkedIn profiles with 7.8K followers; created posts to highlight partner organizations

Projects

Exact Ising Model Simulation | *MIT Beaver Works Summer Institute (BWSI), Q#, Qiskit* June 2025 – Aug. 2025
- Built an exact quantum simulation of the 1D transverse-field Ising spin chain in Q# and Qiskit; published on GitHub
- Applied quantum algorithms (Quantum Fourier Transform, Shor's) to optimize circuit efficiency; presented to 500+ peers

AIDKnIT | *Stanford Clinical Science, Technology and Medicine Internship* June 2024 – July 2024
- Developed AIDKnIT, an AI-driven robotics concept for patient-centered care, with the Stanford medX team and clinicians
- Produced and presented an ePatient case study to clinicians and engineering mentors

SwiftStudy | *HackTJ Hackathon, OpenAI API, AWS* Mar. 2023
- Built SwiftStudy, an app converting uploaded notes into source-linked study questions; placed 1st of 520

Peer-Reviewed Publications

- Von Hoene, E., **Gupta, A.**, Kavak, H. et al. Evaluation of a spatial microsimulation framework for small-area estimation of population health outcomes using the behavioral risk factor surveillance system. *Int J Health Geogr* (2026). doi.org/10.1186/s12942-026-00485-3
- Von Hoene, E., **Gupta, A.**, et al. A Synthetic Population and Small-Area Estimate Dataset for Health Behaviors and Outcomes in the United States. *Nature Scientific Data*.
- **Gupta, A.**, Péter, S. A., et al. Geographically Weighted Surrogate Models for Rapid Small-Area Chronic Disease Estimation. *Applied Geography*.
- Von Hoene, E., **Gupta, A.**, et al. Evaluating the Transferability of a Synthetic Population Generation Approach for Public Health Applications. *Winter Simulation Conference 2025*. doi.org/10.1109/WSC68292.2025.11338931

Technical Skills

Programming & Software: Python, C, C++, Java, R, Q#, Qiskit, HTML/CSS, Git/GitHub
Data & Analytics: Statistical modeling, data visualization (Tableau, Power BI), Google Analytics
Engineering Tools: AutoCAD, Fusion 360, Onshape

Honors & Interests

Honors: Congressional Award Gold Medalist, International Research Olympiad Semifinalist, 1st Place Continental Math League
Interests: Rock Climbing, Baking, Traveling, Anything Dogs